SEQUENCE TO SEQUENCE LEARNING

WITH NEURAL NETWORKS

A foundational encoder–decoder approach using deep LSTMs • NeurIPS 2014

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01 RESEARCH PROBLEM

Many important tasks map variable-length input sequences to variable-length outputs. Standard feedforward networks expect fixed-size representations, making general sequence-to-sequence mapping difficult.

02 OBJECTIVE & MOTIVATION

Develop a general end-to-end neural approach with minimal assumptions about sequence structure. The model must encode an input sequence and generate an output sequence even when their lengths differ.

03 DATASET & EXPERIMENTAL SETUP

- Task: WMT'14 English-to-French translation
- Training subset: 12 million sentence pairs
- Source vocabulary: 160,000 words; target: 80,000
- Deep LSTMs; best direct-translation result uses an ensemble of five LSTMs.

04 CRITICAL OBSERVATION

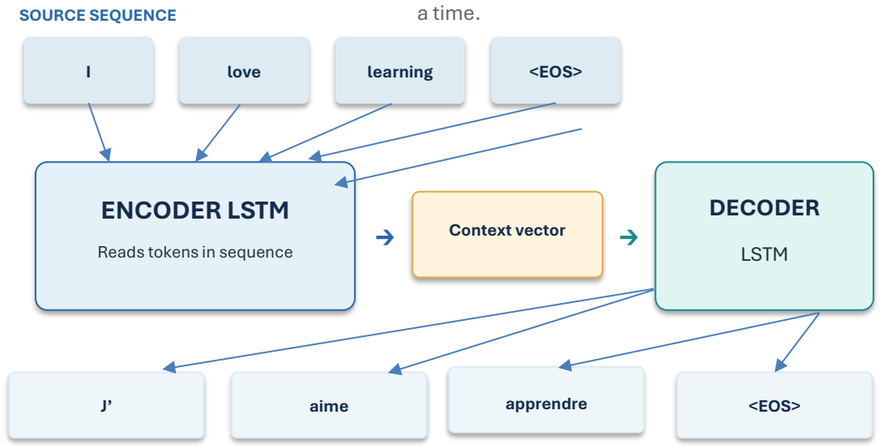
Strength: The paper demonstrates end-to-end sequence learning without requiring a fixed alignment between input and output words.
Limitation: Compressing the whole input into one fixed-length vector can create an information bottleneck. This is a student analysis of the design.

05 CONCLUSION & FUTURE SCOPE

The authors showed that deep LSTMs could outperform a phrase-based statistical MT baseline. A logical future direction is attention, which lets a decoder use information from different input positions rather than relying only on one fixed vector.

06 PROPOSED METHODOLOGY: ENCODER–DECODER

The encoder reads the source sentence; the decoder generates the target sentence one token at a time.



Conceptual illustration: English → French. Input and output lengths can differ; decoding stops at an end-of-sequence token.

07 KEY TECHNICAL IDEAS

Deep LSTMs
Four layers; designed to learn long-range dependencies.

Reverse source order
Improved optimization by creating shorter source–target dependencies.

Beam search
Searches candidate output sequences for a high-probability translation.

Ensemble
Five LSTMs produced the paper's 34.8 BLEU direct-translation result.

08 EXPERIMENTAL RESULTS

34.8 BLEU
Ensemble of 5 LSTMs (direct translation)

33.3 BLEU
Phrase-based SMT baseline

When reranking 1,000 baseline hypotheses, the LSTM ensemble reached 36.5 BLEU. Results are from the paper's WMT'14 evaluation.

09 WHAT THE RESULTS SHOW

- The neural system beat the 33.3 BLEU phrase-based baseline in direct translation (34.8 BLEU).
- Reranking baseline candidates raised the score to 36.5 BLEU.
- Reversing source sentences helped; the paper reports good performance on long sentences.

10 LIMITATIONS & CONTEXT

- A fixed-length context vector may compress information from long inputs.
- Fixed vocabularies were used; out-of-vocabulary words were replaced with UNK.
- Results belong to the 2014 setup and are not current state-of-the-art claims.
- Attention is a later development, not part of this paper's model.

11 KEY TAKEAWAY

One LSTM reads; another writes. This end-to-end design helped establish a powerful framework for mapping sequences to sequences.

12 IMPROVEMENTS

1.Adding Improvement Layer
2. Using Deep LSTM
3. Reverse the Input

13 REFERENCES

[1] I. Sutskever, O. Vinyals, and Q. V. Le, "Sequence to Sequence Learning with Neural Networks," in Advances in Neural Information Processing Systems 27, 2014, pp. 3104–3112. [Online]. Available: https://arxiv.org/abs/1409.3215
[2] D. Bahdanau, K. Cho, and Y. Bengio, "Neural Machine Translation by Jointly Learning to Align and Translate," arXiv:1409.0473, 2014. (Cited as later context for attention.)

Github Link :- https://github.com/ShivamXCode01/Sequence-to-Sequence-Learning-with-Neural-Networks-Research-Paper-